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Iterative Development Model

The **Iterative Development Model** is a process where a large project is broken down into smaller, manageable chunks called **iterations**. Instead of building the whole system at once, developers build it piece by piece.

The Iteration Cycle

Each phase or iteration follows a complete development lifecycle:

1. **Requirement:** Identifying needs for a specific feature.
2. **Design:** Planning the technical architecture of the feature.
3. **Development:** Writing the actual source code.
4. **Testing:** Verifying the feature to find and fix defects.
5. **Deployment:** Releasing the specific feature for use or evaluation.

Agile Testing

1. Definition of Agile Testing

Agile testing is a software testing practice that follows the principles of agile software development. Unlike the Waterfall model, it is an **ongoing process** rather than a separate phase. Testing starts at the beginning of the project and continues throughout development.

2. Agile Testing Key Principles

- **Continuous Testing:** Testing is conducted continuously as the code is developed.
- **Continuous Feedback:** Provides constant feedback to developers to ensure the product meets business requirements.
- **Collaborative Effort:** Testing is a shared responsibility of the entire team (Developers, Testers, and Product Owners).
- **Minimal Documentation:** Focuses on "just enough" documentation instead of lengthy test plans.
- **Test-Driven Development (TDD):** Writing tests before writing the actual code.

3. Agile Testing Methodologies

- **BDD (Behavior Driven Development):** Focuses on the system's behavior from the user's perspective using **Given-When-Then** scenarios.
- **ATDD (Acceptance Test Driven Development):** The team collaborates to define acceptance criteria before development begins.
- **TDD (Test Driven Development):** A development technique where you must first write a test that fails, then write code to pass that test.
- **Exploratory Testing:** A manual testing approach where testers explore the application without pre-defined scripts to find hidden bugs.

4. Benefits of Agile Testing

- **Early Bug Detection:** Since testing is continuous, bugs are found and fixed early.
- **High Product Quality:** Constant feedback ensures the final product is robust.
- **Flexibility:** Easily adapts to changes in requirements during the development cycle.
- **Faster Delivery:** Reduces the time to market by integrating testing with the build process.

Levels of Testing

- Unit Testing
- Integration Testing
- System Testing
 - System testing
 - System integration testing
- User acceptance Testing